

Raw reads

- Obtained sequence data from Novogene.
- This was short read using NovaSeq X Plus Series sequencing strategy which is a next generation sequencing platform.
- Provides with .fq files of each sample site containing all the sequencing data.

FastQC

- Checks the quality of reads and cleans them up into a new file of cleaned reads only.
- Quality of reads is based on high quality scores (q20, Q30), whether sequence adapters are presents (illumina adapters used), how many unidentified bases (N), read lengths.
- Sequences length is 150 as paired-end sequencing PE150 was used.
- Run FastQC again to ensure cleaned reads are improved

MEGAHIT assembly

- Once quality-checked reads MEGAHIT looks for overlaps between the millions of short fragments and joins them together into longer contigs
- fast and memory-efficient de novo assembler designed for large and complex short-read metagenomic datasets and supports multiple k-mer sizes to accommodate uneven sequencing coverage.
- Li D, Liu CM, Luo R, Sadakane K, Lam TW. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. Bioinformatics. 2015 May 15;31(10):1674-6. doi: 10.1093/bioinformatics/btv033. Epub 2015 Jan 20. PMID: 25609793.

Bowtie2 mapping

- This maps my cleaned reads back onto the contigs made by MEGAHIT.
- This shows how much coverage each contig has.
- Bowtie2 was used for mapping as it is well suited to mapping **short, paired-end illumina reads**. It is also ultra-fast and memory-efficient at handling the billions of short reads typical of metagenomic datasets.
- Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. Nat Methods. 2012 Mar 4;9(4):357-9. doi: 10.1038/nmeth.1923. PMID: 22388286; PMCID: PMC3322381.

SAM to BAM

- SAM stands for Sequence Alignment Map. It is a plain text file.
- It needs to be converted to a BAM file so it is compressed and a binary format for downstream analysis
- BAM stands for Binary Alignment Map

Depth

- Check how many reads are mapped across each contig.
- Contig coverage is calculated from the mapped reads to provide depth information for MetaBAT2 binning
- E.g. Contig A has 20x depth

Binning

Table 1 Data Quality Summary

Sample	Library_Flowcell_Lane	Raw reads	Raw data	Effective(%)	Error(%)	Q20(%)	Q30(%)	GC(%)
HS_Biofilm	EKDN250021697-1A_22VGFMLT4_L4	23556768	6.8	98.49	0.01	98.91	95.76	57.81
HS_Biofilm	EKDN250021697-1A_22VGJ7LT4_L8	21839542		98.44	0.01	99.00	96.19	57.83
GB_Biofilm	EKDN250021698-1A_22VGJ7LT4_L8	65122332	9.8	99.04	0.01	99.03	96.27	58.03
HS_Sediment	EKDN250021699-1A_22VGJ7LT4_L4	49818122	7.5	98.70	0.01	99.04	96.08	60.59
GB_Sediment	EKDN250021700-1A_22VGJ7LT4_L4	51114874	7.7	98.87	0.01	99.24	96.76	55.84